Fiscal Year:	FY 2016	**Task Last Updated:**	FY 01/04/2017
PI Name:	Turner, Russell T Ph.D.		
Project Title:	Role of Marrow Adipocytes in Bone Loss during Simulated Spaceflight		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
PI Email:	Russell.Turner@oregonstate.edu	**Fax:**	FY 541 737 6914
PI Organization Type:	UNIVERSITY	**Phone:**	541 737 9545
Organization Name:	Oregon State University		
PI Address 1:	Nutrition & Exercise Sciences/Skeletal Biology Laboratory		
PI Address 2:	Milam Hall, Mail Stop 107B		
PI Web Page:			
City:	Corvallis	**State:**	OR
Zip Code:	97331-8558	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Space Biology NNH11ZTT002N
Start Date:	09/15/2012	**End Date:**	09/14/2016
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
Contact Monitor:	Sato, Kevin	**Contact Phone:**	650-604-1104
Contact Email:	kevin.y.sato@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Extended to 9/14/2016 per A. Chu/ARC and NSSC information (Ed., 7/16/15)		
Key Personnel Changes/Previous PI:	There were no changes in key personnel.		
COI Name (Institution):	Iwaniec, Urszula (Oregon State University) Wong, Carmen (Oregon State University)		
Grant/Contract No.:	NNX12AL24G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Spaceflight, by impairing the differentiation program of stem cells residing within bone marrow results in bone loss, increased bone marrow adiposity, anemia, and impaired immune function. Because these disturbances may potentially compromise the success of long-duration flights, there exists an urgent need to identify the underlying mechanisms and implement effective countermeasures. We propose to test the novel hypothesis that negative energy balance during spaceflight results in 1) bone marrow dysfunction and increased marrow adiposity and 2) altered signaling by peripheral adipokines (cytokines produced by adipocytes) all of which, in turn, contribute to negative bone turnover balance and bone loss. The proposed research will focus on the precise role of adipose tissue and the important adipokine leptin in mediating the detrimental effects of simulated spaceflight (hindlimb unloading (HU)) on bone marrow cells in mice by accomplishing the following 2 specific aims: Specific Aim 1 will test the hypothesis that increased bone marrow adiposity plays a causal role in mediating simulated spaceflight-induced changes in bone metabolism. This will be accomplished by comparing the skeletal response to hindlimb unloading of wild type (WT) mice with mice incapable of generating bone marrow adipocytes (kit-receptor-deficient KitW/W-v mice). Specific Aim 2 will test the hypothesis that the adipokine leptin, by targeting the differentiation of bone marrow stem cells, plays a key role in mediating energy sensitive changes in bone metabolism during simulated spaceflight. This will be accomplished by examining whether transplantation of mesenchymal or hematopoietic stem cells from leptin receptor-deficient db/db mice into WT mice alters bone loss during simulated spaceflight. Completion of the proposed project is expected to increase our understanding of the role of energy metabolism in mediating bone loss during simulated spaceflight and suggest specific non-pharmacological and pharmacological interventions to mitigate the detrimental effects of spaceflight on bone marrow.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Osteoporosis is a major health concern. To our knowledge, all forms of osteoporosis involve infiltration of fat into bone marrow. If the increased marrow fat alters the magnitude of bone loss, as hypothesized, this research will have significance to human health that transcends protecting astronauts from bone loss during spaceflight. Successful completion of the proposed research involved substantial model development and validation and this work will have impact far beyond the specific aims of this project. To accomplish Specific Aim 1, it was necessary to perform adoptive transfer of WT bone marrow cells into kitw/w-v mice (Specific Aim 1). While bone marrow transplantation is routinely performed in the field of immunology, the impact of whole body sub-lethal and lethal radiation typically used to perform the procedure on the skeleton has received little attention. In the course of validating our model we performed several pilot studies that led to the unexpected finding that high dose radiation with photons, while having devastating effects on the hematopoietic compartment actually induces bone lining cells to become osteoblasts, leading to a transient increase in bone formation. Based on the likelihood that this activation of bone lining cells represents a potential therapeutic target we hope to pursue the mechanism for this response. Additionally, we made major advances in optimizing adoptive transfer for bone research: we have developed a protocol where following lethal irradiation we have quantitatively replaced hematopoietic cells in bone marrow with donor cells with no bone loss. This was accomplished with as few as 750 purified hematopoietic stem cells (publication in review and publication in preparation). Mice exhibit premature age-related cancellous bone loss. Specifically, as much as 2/3 of the cancellous bone in the distal femur metaphysis is resorbed prior to establishment of peak total bone mass. The negative implication of this bone loss for HU studies is obvious. The mechanism, however, was unknown. In the course of performing studies in which mice were housed at thermoneutral (32°C) to minimize adaptive thermogenesis, we noted that the mice housed at this warm temperature showed unusually high cancellous bone volume fraction. Further investigation confirmed that premature cancellous bone loss in mice housed at room temperature is caused by mild cold stress (Iwaniec et al., 2016). While the low cancellous bone mass at clinically relevant sites in skeletally mature mice is an issue for preclinical studies, an even more important issue is whether physiological adaptation to mild cold stress influences the outcome of preclinical studies. This is of particular concern to spaceflight studies where mice housed at subthermoneutral temperatures during microgravity are likely to be more dependent upon adaptive thermogenesis than weight bearing controls, leading to bone loss independent of reduced weight bearing. Reference: Iwaniec UT, Philbrick KA, Wong CP, Gordon JL, Kahler-Quesada AM, Olson DA, Branscum AJ, Sargent JL, DeMambro VE, Rosen CJ, Turner RT. Room temperature housing results in premature cancellous bone loss in growing female mice: implications for the mouse as a preclinical model for age-related bone loss. Osteoporos Int. 2016 Oct;27(10):3091-101.
	Specific Aim 1: Test the hypothesis that increased bone marrow adiposity plays a causal role in mediating simulated spaceflight-induced changes in bone metabolism. We evaluated the impact of a 14d spaceflight (in slowly growing ovariectomized rats) and a 14d simulated spaceflight (hindlimb unloading (HU) in adult male mice) on bone marrow adiposity. As described in detail in Keune et al., 2016, bone marrow adipose tissue (MAT) increased in the distal femur metaphysis during spaceflight. During the same interval there was a reduction in cancellous bone volume fraction. The observed bone loss was due to increased bone resorption; osteoblast number, osteoblast activity and osteoblast turnover were normal. In agreement, HU resulted in bone loss in male mice and the bone loss was due to increased bone resorption. However, we were surprised to find that HU did not result in an increase in MAT (publication under review) in WBB6F1/J mice. It should be noted that this unexpected finding was replicated in HU C57BL6 mice (publication in preparation). We conclude from these results that increased MAT is not a prerequisite for bone loss in HU mice. We originally hypothesized that mice incapable of producing MAT would be protected from HU-induced cancellous bone loss whereas mice with excessive MAT would exhibit increased bone loss. We tested these hypotheses using kitw/w-v (MAT-deficient) mice and ob/ob (excessive MAT) mice. As expected, decreased MAT was associated with increased bone formation and increased MAT was associated with decreased bone formation. However, HU-induced cancellous bone loss was exaggerated in MAT- mice and not impacted by excessive MAT (publication in preparation). Taken together, our studies demonstrate that spaceflight-induced cancellous bone loss in ovariectomized rats and HU-induced cancellous bone loss in male mice each occur as a result of increased bone turnover, where the increase in resorption is inadequately compensated for by bone formation. While lineage decision (mesenchymal stem cells to osteoblasts or adipocytes) was shown to influence bone formation and MAT levels, our findings do not support a causal relationship between MAT infiltration and bone loss during spaceflight. Indeed, our studies in MAT-deficient mice suggest that MAT may have a protective role in limiting bone loss. Further research will be required to verify these

Task Progress:	findings and determine the underlying mechanisms. Specific Aim 2: Test the hypothesis that the adipokine leptin, by targeting the differentiation of bone marrow stem cells, plays a key role in mediating energy sensitive changes in bone metabolism during simulated spaceflight. We performed a series of studies to test this hypothesis. Initial studies were focused on establishing the effects of leptin on MAT and bone turnover in normal rats and leptin-deficient ob/ob mice (Turner et al., 2014; Turner et al., 2015; Philbrick KA et al., 2015; Lindenmaier et al., 2016). We also addressed the controversy as to whether the physiological actions of leptin on bone are mediated through peripheral or central nervous system (CNS)-mediated signaling (publication under review). This question is important because leptin regulates energy metabolism through CNS signaling. Subsequent studies focused on how leptin impacts bone and energy metabolism during HU (publication in preparation). Taken together, our studies provide strong evidence that leptin administered subcutaneously and hypothalamic leptin gene therapy have surprisingly similar effects on bone metabolism in ob/ob mice; specifically, leptin decreased MAT, increased osteoblast number and activity, and increased osteoclast activity. Dose response studies revealed that the skeletal effects of leptin occur at very low circulating levels of the hormone (levels having minimal effects on food intake or body weight), suggesting that previously published reports demonstrating skeletal effects of leptin delivered into the hypothalamus were due to transport of leptin into peripheral circulation and not due to CNS mediated actions. If confirmed, our finding that the actions of peripheral leptin on bone and central actions of leptin on energy metabolism are dissociable from one another would represent a paradigm shift. It is well established that there is a close association between the magnitude of mechanical loading of the skeleton and bone mass, and that bone loss often accompanies weight loss induced by dieting (Iwaniec and Turner, 2016). Potentially, this is due to combined decreases in peripheral and CNS leptin levels. In support, we have shown that increasing hypothalamic levels in adult rats can reduce body weight but, in contrast to caloric restriction, promote weight loss without bone loss (Turner et al., 2015). It is well established that weight loss due to reduced food intake contributes to bone loss during HU and for this reason, weight bearing controls are normally pair-fed to HU rodents. However, as shown for weight bearing rodents (Turner et al., 2015), leptin treatment of HU ob/ob mice, although inducing weight loss, did not result in further bone loss (publication in preparation). References Keune, JA, Philbrick KA, Branscum AJ, Iwaniec UT, Turner RT. Spaceflight-induced vertebral bone loss in ovariectomized rats is associated with increased bone marrow adiposity and no change in bone formation. npj Microgravity. 2016 April 2: 16016. Turner RT, Philbrick KA, Wong CP, Olson DA, Branscum AJ, Iwaniec U. Morbid obesity attenuates the skeletal abnormalities associated with leptin deficiency in mice. J Endocrinol. 2014, Oct;223(1):M1-15 Turner RT, Dube M, Branscum AJ, Wong CP, Olson DA, Zhong X, Kweh MF, Larkin IV, Wronski TJ, Rosen CJ, Kalra SP, Iwaniec UT. Hypothalamic leptin gene therapy reduces body weight without accelerating age- related bone loss. J Endocrinol. 2015 Dec;227(3):129-41. Philbrick KA, Turner RT, Branscum AJ, Wong CP, Iwaniec UT. Paradoxical effects of partial leptin deficiency on bone architecture in growing female mice. Anat. Rec. 2015 Dec;298(12):2018-29. Lindenmaier LB, Philbrick KA, Branscum AJ, Kalra SP, Turner RT, Iwaniec UT. Hypothalamic leptin gene therapy reduces bone marrow adiposity in ob/ob mice fed regular and high-fat diets. Front Endocrinol (Lausanne). 2016 Aug 16;7:110. Iwaniec UT, Turner RT. Influence of body weight on bone mass, architecture and turnover. J Endocrinol. 2016 Sep;230(3):R115-30.
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